

A Bayesian optimized spatial-temporal LSTM for traffic flow prediction

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Abstract: Traffic state forecasting plays a vital role in the Intelligent Transportation System, a critical topic in transportation. Researchers have recently begun combining correlated information, such as spatial dependency, with this issue. A Bayesian optimized spatial-temporal attention long short-term memory neural network is proposed to improve the prediction performance. Considering the latent influence of spatial position and temporal period, a hybrid model based on long short-term memory utilizes convolutional modules and attention mechanisms to dig out the spatial and temporal features. Then, to reduce the influence of hyperparameters in the models, Bayesian optimization is utilized to search out the most suitable hyperparameters to make the forecasting performance the best. The RMSE and MAE of BO-STLSTM perform best in the real-world dataset comparison experiments at 25.77 and 18.98, respectively. Furthermore, the experiments also proved that the hybrid model has the best forecasting performance when the number of training epochs is 400.

Key Words: Intelligent Transportation System, traffic volumes forecasting, spatial-temporal attention, Bayesian optimization

1 Introduction

Traffic prediction is a central point of the Intelligent Transportation System (ITS), which forecasts traffic flow volumes, travel traveling times, and traffic running speeds at a specific future time or period [1]. In recent years, more and more research about ITS has been concentrated on accurate and efficient traffic state prediction.

The temporal features of traffic state prediction first come to the researchers' vision fields. AutoRegressive Integrated Moving Averaging (ARIMA), a type of parametric method, is used to predict the traffic flow [2], and the variant of ARIMA, which utilizes seasonal information, seasonal-ARIMA (SARIMA), is proposed in [3]. However, there is an apparent weakness in parametric methods that parametric methods can not handle missing values in big data. When more information, such as spatial correlations, is considered, the parametric methods can not deal with it effectively [4, 5].

Non-parametric methods are adopted to overcome the shortages of parametric methods. Support Vector Regression (SVR), in which the basic idea is to use a linear model to predict in a higher dimensional feature space transformed from the original input space, is widely used in traffic state prediction [6–9]. Another non-parametric method k -Nearest Neighbor (k -NN), which has become commonly used for its simple training process, is introduced to traffic prediction in [10, 11].

Even though the above methods obtain relatively accurate and efficient results from these studies, the profound relationship between data needs to be mined. For this reason, Deep Learning (DL) methods are introduced to the traffic state prediction area and have shown promising performance because they can utilize the correlations among original data. The basic idea of DL is to use deep artificial neural networks (DNN) to match the labels and training data, where no parameters are attached to the original data. However, the DL model's hyper-parameters, such as layers of neural networks and learning rate, are crucial for the result.

Long and Short-Term Memory (LSTM) is an extension of Recurrent Neural Network (RNN), adding memory modules to enable the model to memorize historical inputs, giving this model an excellent performance time-series data. The researches on LSTM application in traffic state prediction have gained many achievements. X. Ma et al. [12] design a three layers LSTM whose hidden layer consists of memory blocks, which can adaptively determine the optimal time lags. W. Wei et al. [14] propose an LSTM-based method with autoEncoder to extract spatial features.

Convolutional Neural Network (CNN), a widely used algorithm in computer vision, is utilized to obtain spatial features. C. Song et al. [15] and X. Ma et al. [16] propose CNN-based models to forecast traffic speeds and have superior performance than their benchmark models. Error-feedback recurrent CNN (eRCNN), learning from prediction errors, is proposed in [17] to utilize the correlations among nearby segments for traffic speed prediction. This model can keep high prediction accuracy on abrupt traffic accidents.

However, the forecasting performance of the DL methods is easily influenced by hyperparameters. To overcome this problem, this paper proposes a state-of-the-art model for traffic volume prediction problems named Bayesian optimized Spatial-Temporal attention LSTM (BO-STLSTM). The main contribution of this paper can be summarized as follows:

- A spatial-temporal attention mechanism is adopted to collect the spatial-temporal features on the target and its upstream and downstream crossroads.
- The Bayesian optimization is adopted to find the best hyperparameters composition.

The remainder of this paper continues as follows. Section 2 presents the related works, such as the definitions of traffic flow prediction and the principle of Bayesian optimization. Section 3 presents the details of the proposed DL algorithm for short-term traffic prediction. In section 4, the forecasting experiments about the comparison with some existing algorithms are carried out. Eventually, the conclusions about the proposed model are presented in section 5.

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2 Related works

2.1 Traffic flow prediction

Definition 1 ([18]) *Traffic network can be regarded as a time-varying graph $\mathcal{G}_t = (\mathcal{V}_t, \varepsilon)$, where $\mathcal{V}_t$ means a finite set of vertices at the t -th time step, which is corresponding to the n sensors set on Traffic network; ε means the edges between n vertices.*

Let x_i^t be the traffic volume of the target vertice i at t -th time step, x_{iu}^t, x_{id}^t are the traffic volume of the upstream and downstream vertices of target vertice i . The x_i^t, x_{iu}^t and x_{id}^t can be represented by a tensor $\mathbf{X}_i^t \in \mathbb{R}^{n \times m}$, where m is the sum of rows of x_i^t, x_{iu}^t and x_{id}^t . Based on the above definition of symbols, the traffic flow prediction can be stated below.

Problem 1 ([19]) *Give a p -length time slot historical traffic flow tensor of the target vertice i at t -th time step $\mathbf{X} = \{\mathbf{X}_i^{t-p+1}, \dots, \mathbf{X}_i^{t-1}, \mathbf{X}_i^t\}$, the objective is to obtain the traffic flow prediction after q -length time slot future traffic flow tensor $\mathbf{X}' = \{\mathbf{X}_i^{t+1}, \dots, \mathbf{X}_i^{t+q-1}, \mathbf{X}_i^{t+q}\}$.*

2.2 LSTM

Long short-term memory (LSTM) [20] is widely used in dealing with time-series problems, such as machine translation, speech recognition, and music generation. The computational process in LSTM can be divided into four parts: forgetting process, remembering process, hidden cell updating process, and output process. The structure of LSTM and the four computational processes are shown in Fig.1

When at t -th time step, a sigmoid layer, which is named forget gate, firstly obtain the output of forget gate f_t by combining the output of the last time step h_{t-1} and the input of the current time step x_t , and then combine f_t with the hidden layer state of last time step c_{t-1} . The process of forget gate f_t can be written as Eq.1.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \quad (1)$$

where W_f and b_f are the weight and bias of forget gate respectively.

Then, the input gate determines what will be remembered in LSTM by using a sigmoid layer and a tanh layer to deal with h_{t-1} and x_t , and obtains the output of input gate i_t and candidate hidden layer state $\tilde{C}_t$. This process can be written as Eq.2,3.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \quad (2)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C), \quad (3)$$

where W_i and b_i are the weight and bias of input gate respectively, W_C and b_C are the weight and bias of candidate hidden cell layer state respectively.

Next, c_{t-1} with f_t, i_t , and $\tilde{C}_t$ is utilized to obtain the hidden layer state of current time step c_t , which is presented in Eq.4.

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t. \quad (4)$$

At last, the candidate output o_t , which is produced by a sigmoid layer named output gate, combines with the tanh-

processed h_t to obtain the output of LSTM. The last process is displayed as Eq.5, 6.

$$o_t = \sigma(W_o [h_{t-1}, x_t] + b_o), \quad (5)$$

$$h_t = o_t * \tanh(C_t), \quad (6)$$

where W_o and b_o are the weight and bias of output gate respectively.

2.3 Bayesian optimization

J. Ji et al. [21] consider Bayesian optimization (BO) better tuning hyperparameters when the objective function is unknown and expensive to evaluate than the convex objective function in many optimization problem settings. The basic methodology of BO is using Bayes' theorem to the posterior distribution of the objective function and then selecting the hyper-parameters according to the distribution. BO can be divided into exploration and exploitation steps. Exploration means to sample where the global optimal solution is most likely. While exploitation means to sample where has not been sampled. Furthermore, a better-balanced exploration and exploitation will obtain a better global optimum.

Let expected improvement (EI) be the acquisition function to present the tradeoff between exploration and exploitation, obtained by Eq.7 and Eq.8 in mathematical.

$$EI(x) = \begin{cases} (\mu(x) - f(x^+)) \Phi(z) + \sigma(x) \Phi(z) & , \sigma > 0 \\ 0 & , \sigma = 0 \end{cases}, \quad (7)$$

$$z = \frac{\mu(x) - f(x^+)}{\sigma}, \quad (8)$$

where $\Phi(\cdot)$ is the normal cumulative distribution function, $\Phi(\cdot)$ is the normal probability density function, and $f(x^+)$ is the existing maximum value, μ is the mean value, and σ is the variance.

3 Proposed model

Inspired by spatial-temporal attention which is able to extract the features hidden in the high-dimension input data, this paper introduces spatial-temporal attention into LSTM for more accurate traffic volume forecasting, named STLSTM. Furthermore, BO is applied to ensure the hyperparameter choice is optimal, extending the proposed model as BO-LSTM. The overall framework is presented in Fig.2.

Firstly, STLSTM is proposed to extract spatial and temporal features hidden in the input sequence. The spatial and temporal attention can be described as

$$S = V_s \cdot \sigma\{(X \cdot W_1) \cdot W_2 \cdot (W_3 \cdot X)^T + b_s\}, \quad (9)$$

$$E = V_e \cdot \sigma\{(X^T \cdot U_1) \cdot U_2 \cdot (X \cdot U_3) + b_e\}, \quad (10)$$

where X is the input sequence, T is the time step, V_s and $V_e \in \mathbb{R}^{T \times T}$ are attention coefficient, respectively, W_i and $V_i, i \in \{1, 2, 3\}$ are the weights to connect the channels in X , respectively. Regarding spatial features, the traffic volume of the upstream and downstream crossroads simultaneously are utilized. Moreover, in terms of temporal features, we divide the period into three parts: hour-period part, day-period

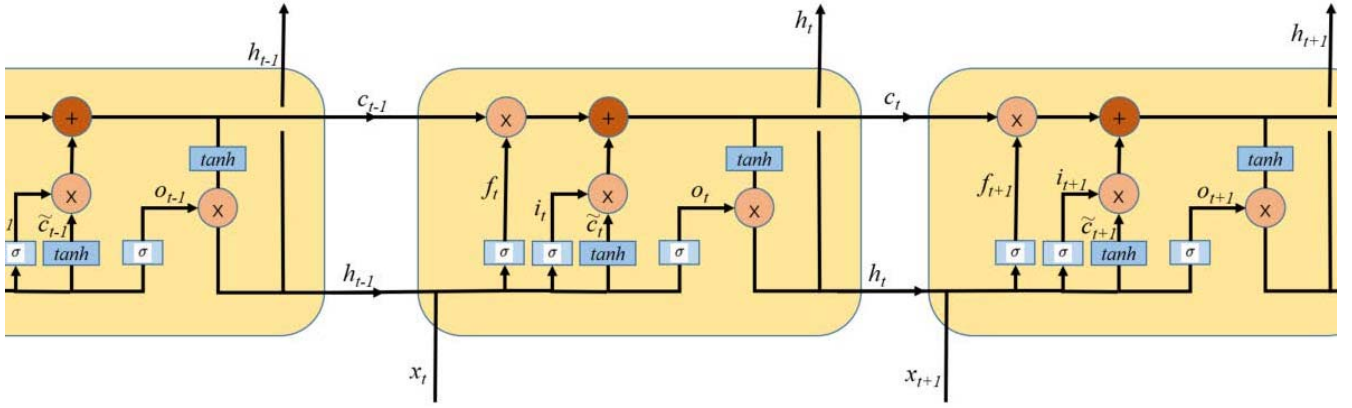


Fig. 1: The diagrammatic drawing of the structure of LSTM

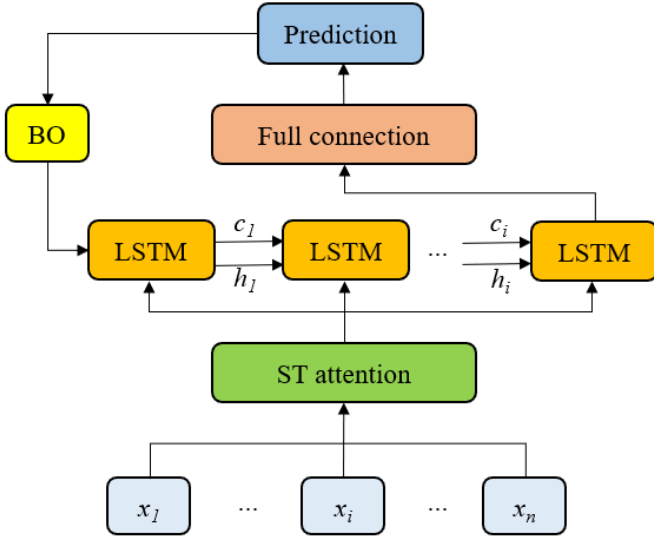


Fig. 2: The structure of BO-STLSTM model.

part, and week-period part, where all the parts contain the last 1-hour traffic data. Secondly, to reduce the influence of the hyperparameters, BO is applied in our model. As an optimization method, BO has a great performance in global searching. The numerical experiments proves the effectiveness of BO in Set.4. The complete process of BO-STLSTM is presented in Algorithm 1.

4 Experiments

In this section, all the experiments are carried out in the python3.6 environment on a 3.1 GHz PC with process I9-9820X CPU, 128 GB RAM, and RTX 2080Ti.

4.1 Case description and parameters setting

The highway dataset collected in California, named PeMS, is applied in this paper to validate the prediction performance of our proposed model.

PeMS datasets, collected by the Caltrans Performance Measurement System [22], have over 39,000 sensors to record the highway traffic conditions in major Californian

Algorithm 1: Framework of BO-STLSTM.

Input: Input traffic volume sequence X .

Output: The predicted traffic volume X_p .

Format the data as time sequence as

$(X_{target}, X_{up}, X_{down})$, and each X consists of x_h, x_d , and x_w ;

Split the training set, validation set, and test set;

Normalize the training sequence.

while Bo is not converged **do**

 Generalize the learning rate and weight decay by BO;

foreach training time $\in$ epochs **do**

 Train the STLSTM model with training set;

 Evaluate the STLSTM model with validation set;

 Output the optimized learning rate and weight decay ;

 Predict the traffic volume with test set.

Table 1: The parameters setting of STLSTM

	Training set	Validation set	Test set
Number	11578	1699	1699

metropolitan areas and aggregate the data into every 5-minutes interval from the original 30-seconds data. In detail, PeMS04, used in this paper, contains 3848 sensors on 29 roads in San Francisco Bay Area from January to February 2018. This paper selects the No.57 sensor as the target sensor, with the upstream and downsrtream sensor as No.118 and No.207, respectively. Some historical data in PeMS04 is shown in Fig.3.

The dataset splitting and parameters setting of STLSTM are shown in Table 1 and Table2, respectively.

4.2 Evaluation indexes and baseline model

In traffic volume prediction, the statistical indices shown below are adopted to measure the performance of forecast models.

- Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |x_{i_p} - x_{i_r}|; \quad (11)$$

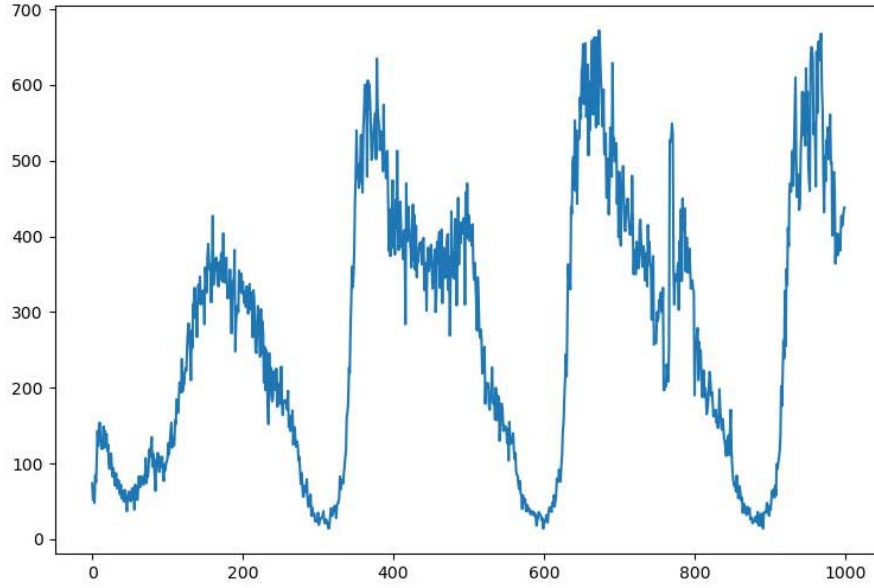


Fig. 3: The first 1000 data of No.57 sensor.

Table 2: The parameters setting of STLSTM

Parameter	Value
Input channels	3
Hidden layers channels	70
Output channels	1
Convolutional kernel size	2 * 2
Global dropout probability	0.6
Learning rate	0.002
Weight decay	0.0004

Table 3: Performance comparison of different models on PeMS04.

Prediction model	RMSE	MAE
HA	54.14	36.76
ARIMA	68.13	32.11
VAR	51.73	33.76
LSTM	45.82	29.45
GRU	45.11	28.65
STLSTM (ours)	31.30	23.36
BO-STLSTM (ours)	25.77	18.98

- Mean Square Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (x_{i_p} - x_{i_r})^2; \quad (12)$$

- Root Mean Square Error (RMSE)

$$RMSE = \left[\frac{1}{n} \sum_{i=1}^n (x_{i_p} - x_{i_r})^2 \right]^{\frac{1}{2}}; \quad (13)$$

where i and n are the sequence index and the length of the whole sequence, respectively. x_{i_p} , x_{i_r} are the volume of prediction and real traffic flow, respectively.

4.3 Preprocessing

All the missing data in datasets are filled by the average volume of the last and next timesteps. In addition, due to the characteristic that LSTM is sensitive to the input data and the requirements of neural network regularization, the data are transformed in zero-mean normalization, defined as Eq.14.

$$x'_i = x_i - \frac{1}{n} \sum_{i=1}^n x_i, \quad (14)$$

where x is the original data sequence, x' is the zero-mean normalized data sequence, i and n are the index and the length of the sequence respectively.

4.4 Baselines

The baselines listed below are compared with the proposed model:

- Historical Average method (HA).
- Auto-Regressive Integrated Moving Average method (ARIMA) [23].
- Vector Auto-Regressive (VAR) [24].
- Long Short-Term Memory network (LSTM) [20].
- Gated Recurrent Unit network (GRU) [25].

4.5 Experiments results

The forecasting performance of STLSTM and BO-STLSTM is shown in Fig.4, and the forecasting performance comparison is listed in Table 3.

From Fig.4, it can be observed that, in terms of forecasting trends, both of the proposed models have outstanding forecasting performance to obtain the rise and decline features, which reflect on the pick and nadir volumes. Regarding forecasting accuracy, the BO-STLSTM has better performance

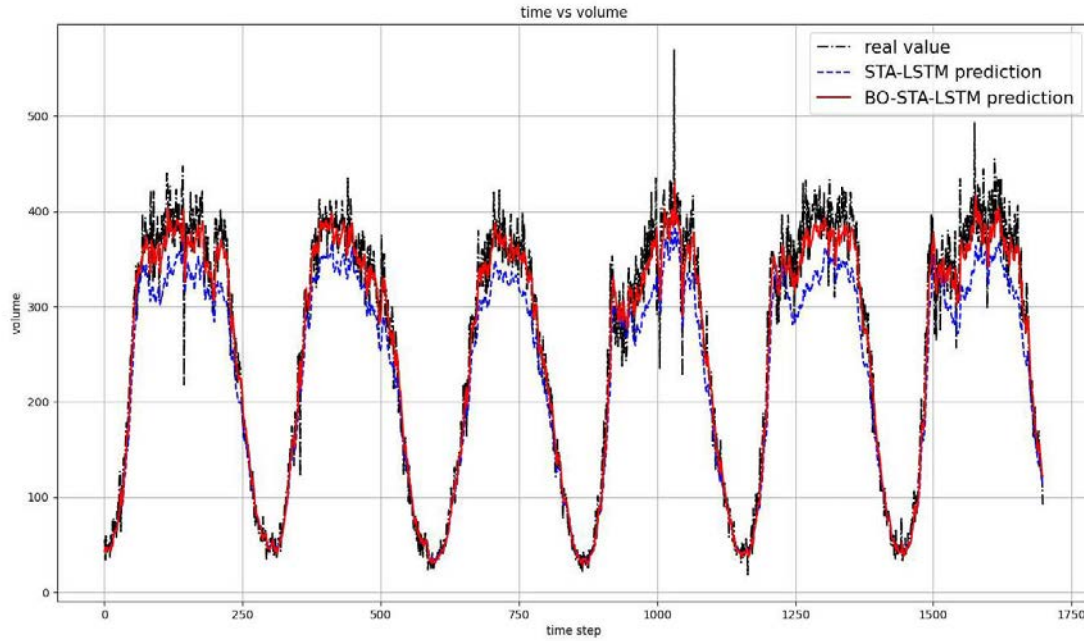


Fig. 4: Forecasting results of STA-LSTM and BO-STA-LSTM.

than STLSTM, considering that more appropriate hyperparameters have been selected.

From Table 3, it is easy to confirm that the proposed BO-STA-LSTM has the best forecasting performance among the seven models in terms of both evaluation metrics. It can be observed that the traditional time series analysis methods such as HA, ARIMA, and VAR could be better for modeling the traffic flow data, which are recognized as complex and nonlinear data. LSTM and GRU, the single deep learning methods, perform better than the traditional time series analysis methods. With spatial and temporal correlation ships introduced into deep learning methods, the forecasting performance obtains extensive improvements. Moreover, BO-STLSTM reduces by 5.78 in RMSE and 4.59 in MAE. In other words, 18.46% and 19.64% reduction in RMSE and MAE, respectively.

Furthermore, the best number of training epochs is researched as well. The forecasting results are recorded every 100 epochs in Fig.5. From Fig.5, it is clear that the minimum coordinates of RMSE and MAE curves are tabbed as 400. When the training epoch is 400, the RMSE and MAE can be reduced to 25.2562 and 18.4995, respectively.

5 Conclusions

In this paper, a novel hybrid model named BO-STLSTM is proposed and applied successfully to predict traffic volumes. First, the STLSTM model utilizes the attention mechanism and convolution layers to simultaneously learn the dynamic spatial-temporal characteristics of traffic data. Then, BO is introduced in STLSTM to reduce the forecasting performance to optimize the hyperparameters. Experiments on the PeMS04 dataset suggest that the forecasting accuracy of BO-STSTM is superior to existing models. Furthermore, the relationships between training epochs and forecasting results

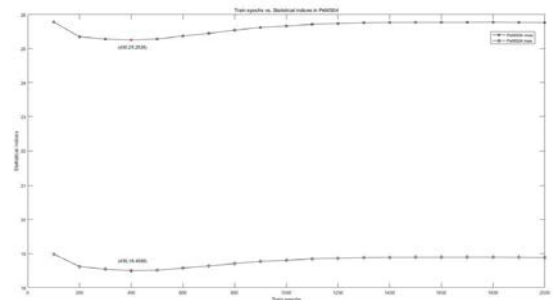


Fig. 5: The statistical indices changes with training epochs in PeMS04 dataset.

are also researched.

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